

Aurora-Lens and Related Architecture: A Prior Art and Precedence Record

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Abstract

This document constitutes a formal prior art and precedence record for the Aurora-Lens governance architecture and its foundational specifications. It establishes the earliest dated public artifacts, the five Australian provisional patent filings, and a structural comparison with subsequently published work in the same architectural space. The record is produced to support patent conversion, licensing, and acquisition discussions, and to provide a citable reference for the chronological and technical priority of this work.

The architecture described here — deterministic runtime admissibility gating, co-produced forensic audit at the moment of governance decision, refusal as a first-class outcome, and hash-chained tamper-evident audit trails — was running in production and subject to patent application in November 2025. This predates all known public publications in adjacent territory by a minimum of four months.

1. Architecture Overview

Aurora-Lens is a provider-agnostic runtime governance proxy that operates at the transport layer between applications and large language model providers. It does not modify underlying models. It enforces deterministic admissibility control before model output becomes consequence.

The architecture comprises five foundational specifications:

- Aurora — the reasoning substrate and admissibility framework
- Persistent Existence Framework (PEF) — the non-temporal persistence substrate on which discourse-level meaning depends
- Aurora+PEF Integration — the integrated governance model combining admissibility control with persistent state
- Compositional Reasoning Architecture (Spine) — the formal reasoning model governing inference under uncertainty
- Autonomous Compositional Reasoning Architecture — the extension to autonomous and agentic execution contexts

The core architectural claims are:

- Governance is a pre-execution admissibility function, not a post-hoc monitoring or compliance process
- Refusal and suspended judgment are valid first-class governed outcomes, not failure modes
- The forensic audit record is co-produced at the moment of governance decision, not reconstructed afterward
- The audit trail is hash-chained with HMAC signatures, making it tamper-evident and deterministically replayable
- The governance layer is non-bypassable by architectural position, not policy assertion
- Session state enables cross-turn contradiction detection and commitment drift detection — the only known governance layer with this property
- The system enforces pathway routing across four outcome classes: PASS, CONTAIN (FORCE_REVISE), HARD_STOP, and ASK

2. Chronological Priority Record

2.1 Patent Filings

Five Australian provisional patent applications were filed between 27 November and 19 December 2025. These establish the earliest formal priority dates for the architecture.

Filing Date	Application Number	Reference
27 November 2025	2025905835	AMCZ-2515186926
28 November 2025	2025905860	AMCZ-2515190345
29 November 2025	2025905885	AMCZ-2515193769
08 December 2025	2025906132	AMCZ-2515223406
19 December 2025	2025906680	AMCZ-2515271247

Non-provisional conversion deadline: November-December 2026. PCT filing in progress.

2.2 Public Record Chronology

Date	Event
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20 November 2025	Conceptual groundwork made public
27 November 2025	Aurora + PEF repository made public; aurora_cli publicly available
6 December 2025	Direct circulation to named researchers in AI governance and alignment
27 Nov - 12 Dec 2025	Heavy cloning and uptake activity documented against public repository
12 December 2025	Repository withdrawn due to cloning without engagement
20 February 2026	Aurora-Lens Demo v1.0 released on Zenodo (DOI: 10.5281/zenodo.18923302 [milarien/aurora-governor-demo])
21 February 2026	OECD Alignment report published on Zenodo (DOI: 10.5281/zenodo.18719033)
19 March 2026	Aurora-Lens v2.0.0 published on Zenodo (DOI: 10.5281/zenodo.18923302)
25 March 2026	Nippon Life v. OpenAI architectural response published on Zenodo
11 April 2026	Epistemic Legitimacy preprint v2 published on Zenodo (DOI: 10.5281/zenodo.19504665)

3. Published Research Record

The following Zenodo records constitute the published research corpus associated with this architecture. All are authored by Margaret Stokes and carry Zenodo-assigned DOIs with timestamped upload records.

Record	Description
DOI: 10.5281/zenodo.19504665 (April 11, 2026)	Epistemic Legitimacy as a Governance Layer for Large Language Models: Architecture and Implementation. Describes Aurora, PEF, Lens, and Governor as distinct architectural components.
DOI: 10.5281/zenodo.18976303 (March 12, 2026)	Present-Centered Cognition Model (PCCM): A Cross-Substrate Framework. Establishes the ontological foundations for present-bound admissibility.
DOI: 10.5281/zenodo.18719033 (February 21, 2026)	Operational Alignment of Aurora-Lens with OECD Due Diligence Guidance for Responsible AI (2026). Maps the architecture to regulatory requirements.
DOI: 10.5281/zenodo.18653120 (earlier version)	Epistemic Legitimacy as a Governance Layer for Large Language Models (v1). First full architectural exposition.

4. Structural Comparison with Subsequent Published Work

In March and April 2026, a body of work describing architecturally convergent runtime governance frameworks appeared on Zenodo and arXiv, published under the authorship of John M. Willis (Sustainable Future Tech, Inc., Washington DC). The primary synthesis paper is: Willis, J. M. (2026). Runtime Execution Governance for AI Systems: A Cross-Platform Synthesis and Architectural Framework. DOI: 10.5281/zenodo.19341177. Published March 30, 2026.

The following table documents the structural correspondence between the Aurora-Lens architecture (earliest priority date: 27 November 2025) and the Willis framework (earliest public record: March 29, 2026).

Architectural Element	Aurora-Lens (Nov 2025)	Willis / AGCP (Mar 2026)
Pre-execution admissibility gate	Lens: runtime admissibility gate evaluated before output becomes consequence	AGCP execution-bound authorization evaluated prior to state transition commit
Append-only tamper-evident audit trail	Hash-chained HMAC audit record co-produced at governance decision time	Append-only ledger with deterministic lifecycle derivation
Non-bypassability as architectural property	Non-bypassability enforced by pipeline position, not policy assertion	Non-bypassability as structural guarantee of control plane
Refusal as first-class outcome	Refusal is a governed terminal outcome, not a failure mode; post-refusal continuation is itself constrained	Deny / escalate as explicit decision outcomes in governance function
Deterministic evaluation semantics	Deterministic policy evaluation with replayable decision states	Deterministic evaluation: identical inputs produce identical outcomes
Co-produced proof at decision time	Forensic envelope generated at the moment of governance decision, not reconstructed post-hoc	Evidence produced at execution point; not reconstructed after the fact
Persistent state substrate	PEF: Persistent Existence Framework as non-temporal state substrate for discourse continuity	Lifecycle state derived from append-only ledger; no independently mutable status fields

The Willis framework does not cite Aurora-Lens, the Aurora specifications, or any work by Margaret Stokes. No independent basis for the convergence has been established by the Willis publications.

A further convergent work, Mazzocchi (2026), Cryptographic Runtime Governance for Autonomous AI Systems: The Aegis Architecture (DOI: 10.5281/zenodo.19027190), is cited within the Willis synthesis, indicating at least three bodies of work converging on the same architectural space in early 2026, all postdating the Aurora-Lens priority record.

5. Production Evidence

Aurora-Lens has been running in production since November 2025. The following constitutes evidence of active deployment predating all published work in the adjacent space.

- Hash-chained audit log with HMAC-signed governance events including PASS, CONTAIN, and HARD_STOP decisions with forensic envelopes. Dated records exist from November 27, 2025 and earlier.
 - PyPI package `aurora-lens`, withdrawn after extraction-without-engagement pattern was identified. Version 2.1.2 was current at withdrawal.
 - Live governance demo at milamba.com/demo.html, publicly accessible.
 - Tracking data documenting engagement with milamba.com/aurora-lens.html, system-capabilities page, evidence page, and PyPI from Cisco Umbrella enterprise network (146.112.x.x) within minutes of March 20, 2026 outreach emails.
 - IBM network (9.169.121.179) click-through to GitHub repository documented December 6, 2025.
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6. Patent Status and Conversion

Five Australian provisional patent applications cover the core architectural claims, filed 27 November through 19 December 2025. Confirmed application numbers include 2025905835 and 2025905860.

Non-provisional conversion deadline falls November-December 2026. PCT filing is in progress. Conversion is under active consideration and subject to funding and strategic engagement.

The patent notice on the Willis AGCP specification repository (github.com/jwillisSFT/agcp-spec) — published March 2026 — references pending patent applications. The Aurora-Lens provisional filings predate this notice by a minimum of 99 days on the earliest provisional (27 November 2025 vs. 29 March 2026).

7. Distinguishing Properties

The following architectural properties of Aurora-Lens are not present in any known prior or concurrent published work and constitute the basis for the strongest novelty claims:

- Session-awareness enabling cross-turn contradiction detection and commitment drift detection across a multi-turn conversation. No other known governance layer operates at this level.

- Co-produced forensic proof at the moment of bind: the audit record is generated at the governance decision point, not reconstructed from logs after the fact. This satisfies a 'could have been interrupted' test that post-hoc audit architectures cannot.
 - Refusal as a lawfully constrained terminal outcome with its own continuation governance via Governor. Post-refusal continuation is itself subject to admissibility constraints.
 - Provider-agnostic transport-layer position: Aurora-Lens operates at the HTTP transport layer, requiring no code changes, no model modification, and a single YAML configuration file.
 - PEF as non-temporal persistence substrate: persistent state is maintained as an existence frame, not a memory sequence, enabling discourse-level governance across interactions without temporal dependency.
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8. Contact and Engagement

Margaret Stokes is open to licensing, pilot, and acquisition discussions with organisations deploying AI in regulated or high-liability environments.

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Precedence record: milamba.com/writing/precedence.html

GitHub: github.com/milarien

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This document is intended as a citable prior art record and does not constitute legal advice.